Setting Up Pi-hole

Recommended Setup

- Raspberry Pi Zero 2 W + 16-32GB microSD card (larger capacity is better for longer use)
- Power Requirement: Minimum 5V/1A, but use a 5V/2.5A or 5V/3A micro USB power supply (like the official Raspberry Pi power supply) to ensure stability for 24/7 operation

Tutorial Resources:

- https://docs.pi-hole.net/main/basic-install/
- Block Ads ANYWHERE You Go — The Pi-hole + Tailscale Setup Nobody Talks About https://youtu.be/LXIL4qxN3lE?si=PiKsFBnljlhWbptp
- Block Ads on ALL DEVICES (Smart TVs) — Simplest Pi-hole Tutorial https://youtu.be/d_3h5n9mPdI?si=QddwtIa29XbqV_6k

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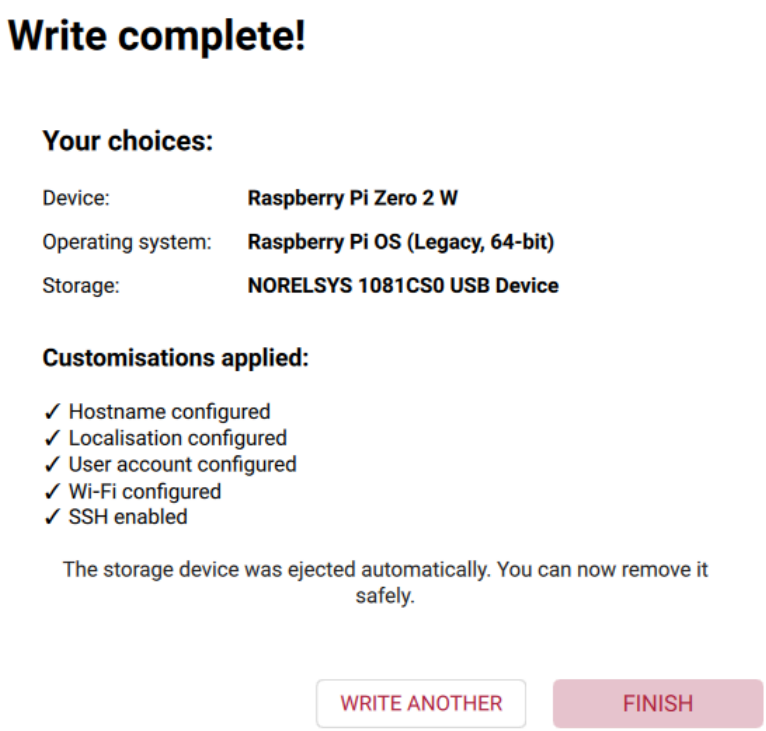
Installation Steps:

In Raspberry Pi Imager, download **RASPBERRY PI OS (LEGACY, 64-BIT) LITE** to the microSD card.

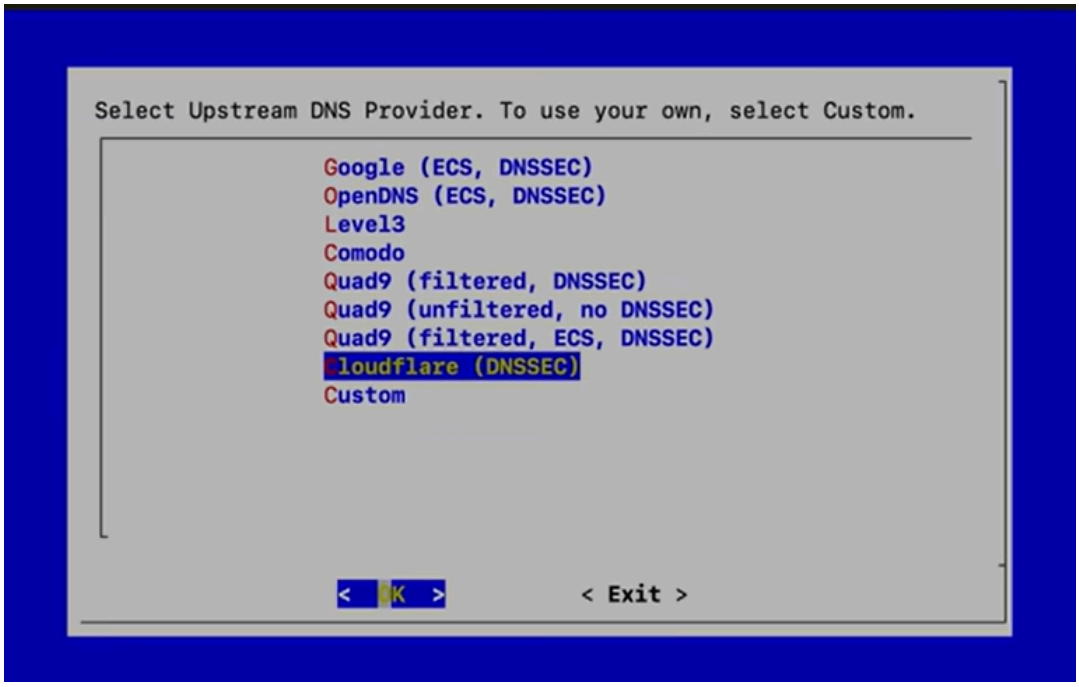
a. Make sure to set your username, hostname, and password. Make sure to set the timezone correctly. Also input your Wifi name and password, so later on the Raspberry Pi can get its IP when its booted up



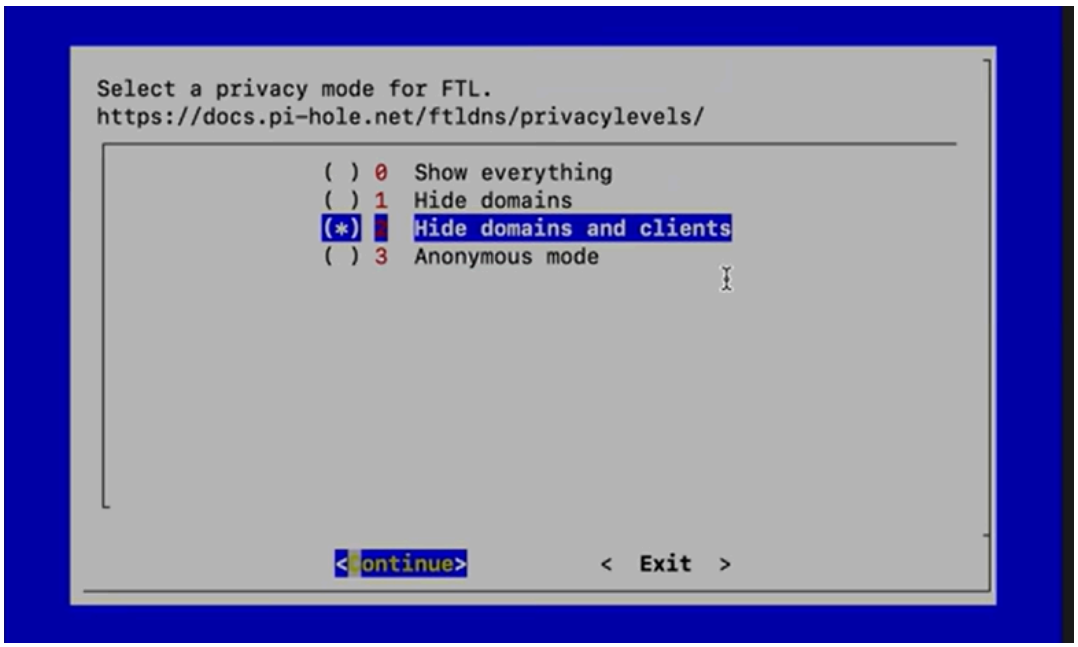
b. Once Pi-hole boots up (LED light shows), SSH from to that Raspberry Pi 2 W using: `username@hostname`, then enter your password
   - In terminal, type: `curl -sSL https://install.pi-hole.net | bash`

c. Follow the YouTube tutorial until the "set up static IP" part, then go to your router admin website , type **192.168.1.1** on Google, then go to DHCP to reserve an IP for your Pi-hole
   i. In case you do not know the router credentials, check the router's back or bottom for admin credentials if needed

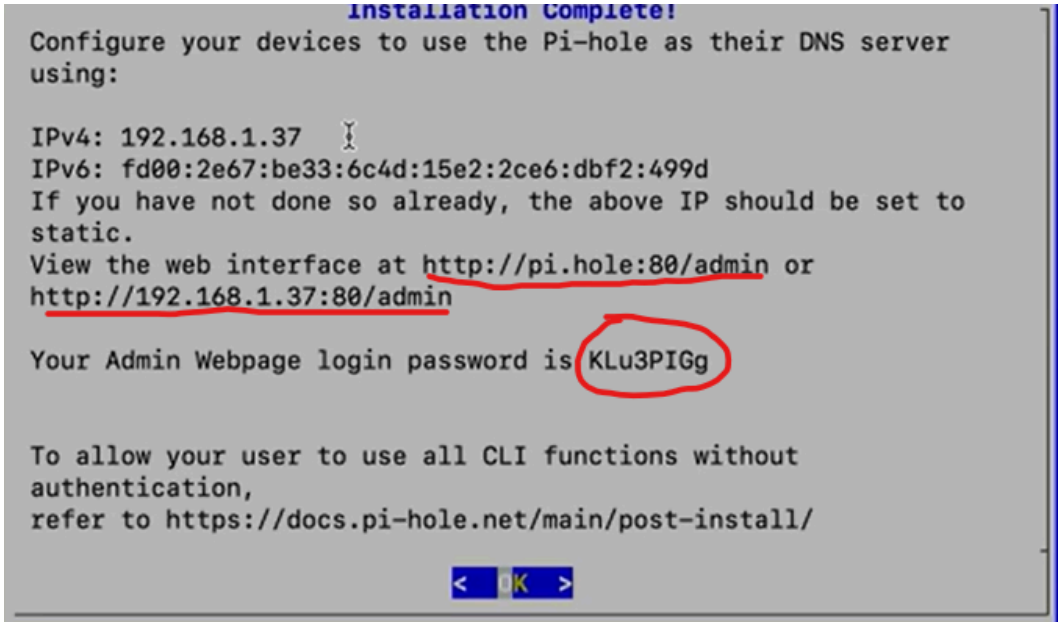
d. Choose Cloudflare as the DNS server for Pi-hole. Cloudflare DNS (1.1.1.1) is generally superior to Google DNS (8.8.8.8) due to faster query speeds, enhanced privacy, and stronger built-in security features



e. Then enter Yes for the next request
f. Select **Hide Domains and Clients** as a privacy mode

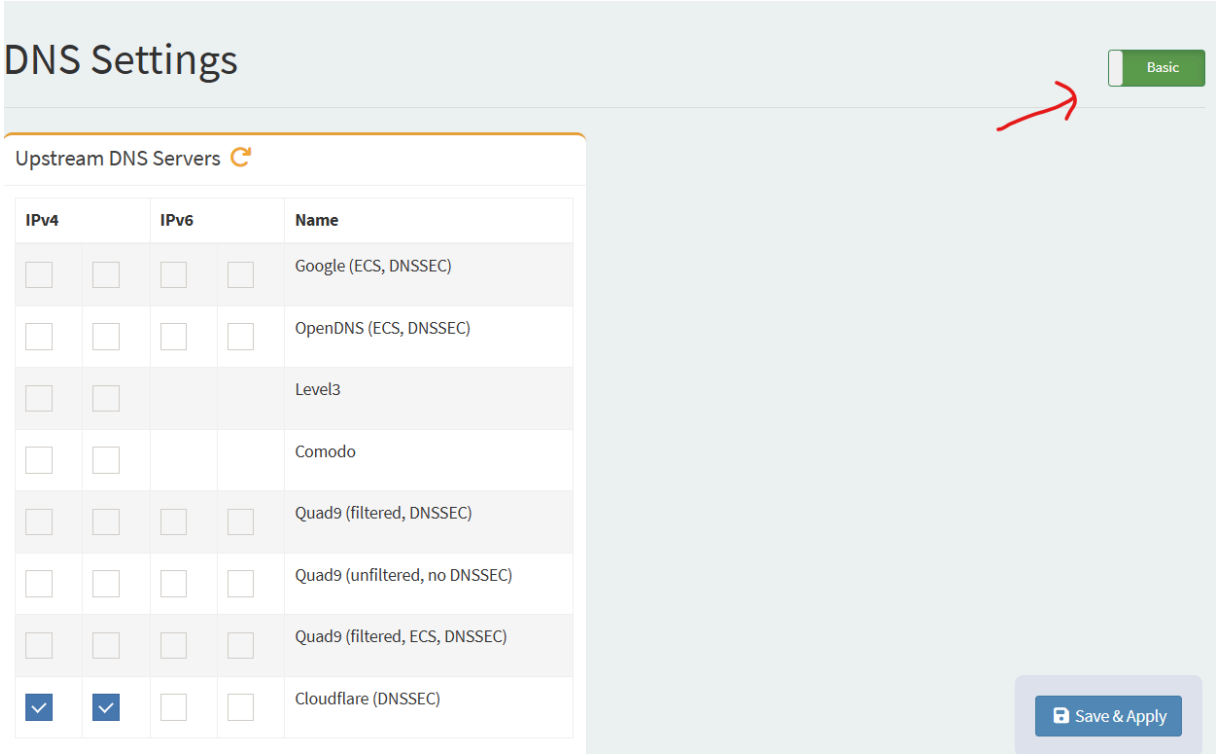


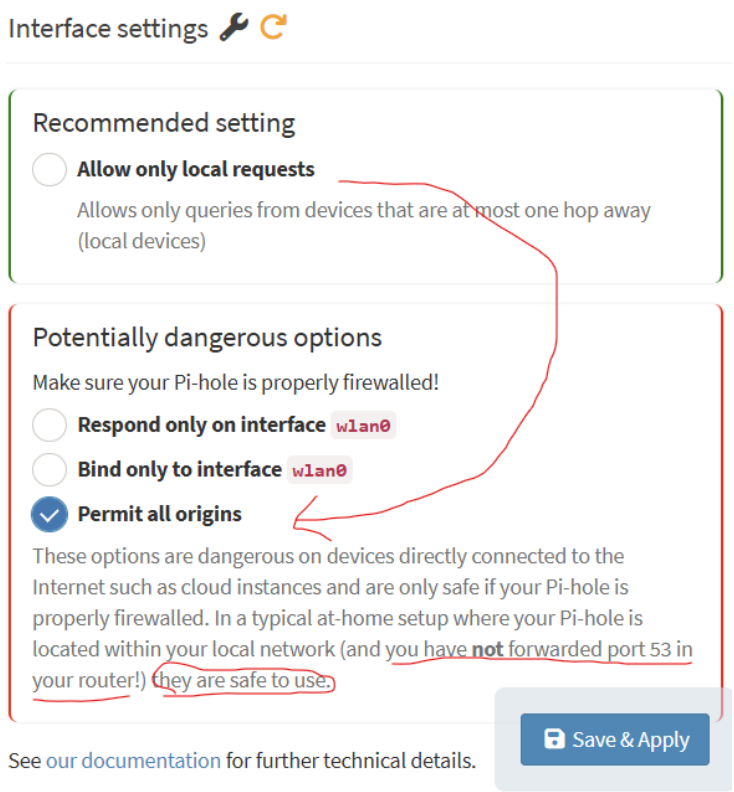
g. After that enter OK and you will lead to this page, make sure to copy one of the URL and login with the password as shown in your page, you can change the password later, using command:
    i. Sudo pihole setpassword



After logging into Pi-hole

In the Pi-hole web interface, go to Settings → DNS → change from Basic to Expert, then select Permit all origin.

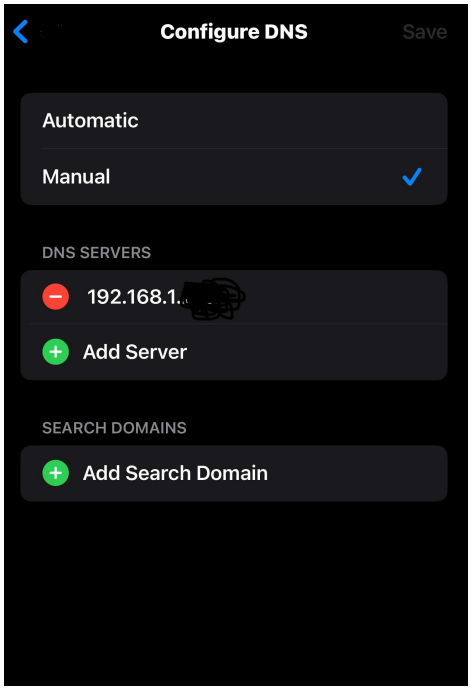




Configuring DNS on Devices

For iPhone:

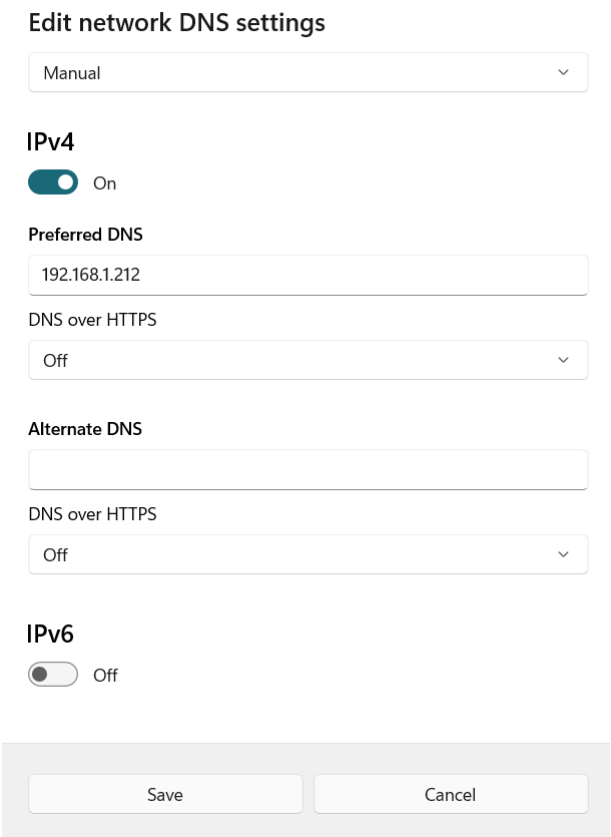
1. Go to Settings → Wi-Fi → click on your current Wi-Fi network
2. Scroll down to DNS (Configure DNS) → change Automatic to Manual
3. Click Add Server → enter your Pi-hole's IP
4. Remove the old DNS server (192.168.1.1) - there must only be one DNS server
5. Done



For Windows:

1. Go to Settings → Network & Internet → Wi-Fi → click on your current Wi-Fi network
2. Scroll down to DNS server assignment → click Edit
3. Change Automatic to Manual
4. Enable IPv4 and disable IPv6 (if not used in Pi-hole)
5. In Preferred DNS, add your Pi-hole's IP and leave Alternate DNS blank
6. Disable DNS over HTTPS option
7. Save & Done

Here is on Window



Pi-hole Troubleshooting Summary

1. Initial Diagnosis:

bash

```
sudo pihole -d
```

Generates debug report showing errors, system health, and configuration.

2. DNS Connectivity Tests:

bash

```
# Test upstream DNS
ping -c 4 1.1.1.1
dig @1.1.1.1 google.com
```

```
# Test TCP specifically (fallback method)
dig +tcp @1.1.1.1 google.com
```

3. WiFi Stability Check:

```bash
# Test packet loss and latency to router
ping -c 100 192.168.1.1

# Check for WiFi driver issues
sudo journalctl -u wpa_supplicant --since "1 hour ago"
```

4. Channel Analysis:

```bash
# See which SSID you're connected to
iwconfig wlan0 | grep ESSID

# Scan all nearby networks and their channels
sudo iwlist wlan0 scan | grep -E "ESSID|Channel"
```

WiFi & Channel Concepts:

2.4GHz Channels:

- Only 3 non-overlapping channels: **1, 6, 11**
- Channels 2-5 overlap with 1 and 6
- Channels 7-10 overlap with 6 and 11
- **Best practice:** Use only 1, 6, or 11

Channel Congestion:

- Multiple networks on same channel = interference
- Causes: packet loss, high latency, connection drops
- **Solution:** Switch to least-used channel (1, 6, or 11)

Multiple SSIDs on One Router:

- Remember if you set up Two SSIDs on the same router = it will use **same physical radio** = make sure to set both SSIDs to the same channel
- Cannot broadcast on different channels simultaneously
- Both move together when you change channel